

Class - 02 (16.02.2026)

Classifier $\rightarrow$ function $\rightarrow h(x) \rightarrow y$
 $\downarrow$ $\rightarrow$ category label
feature
representation of some object

$$h: O \rightarrow Y$$

$$O \rightarrow Y$$

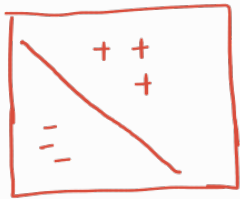
a function that takes an element from set O and outputs an element in set Y .

Classifier has 2 type.

1. Generative classifier $\rightarrow$ Naive Bayes. $P(x, y) = P(x|y)P(y)$

2. Discriminative classifier $\rightarrow$ Logistic Regression

$$P(y|x)$$



decision boundary

X matrix

$$\begin{matrix} f_1(o_1) & f_2(o_2) & \dots & f_m(o_1) \\ \vdots & \vdots & & \vdots \\ f_1(o_n) & f_2(o_n) & \dots & f_m(o_n) \end{matrix}$$

Y vector

$$\begin{matrix} y_1 \\ \vdots \\ y_n \end{matrix}$$

feature vector:

$$x = [f_1(o), f_2(o), \dots, f_m(o)]$$

Generative Classifier:

models how input is generated?

means: If the class is fixed, what kind of feature patterns do we expect?

example:

Instead of thinking:

Given this email, is it spam?

Generative thinking says:

Suppose I tell you this email IS spam.

What kinds of words would you expect to see?

That expectation is:

$$P(x \mid y = \text{spam})$$

I can actually generate fake samples

Discriminative:

directly learns boundary

Bow

→ Bow
→ Bag of Word → Numerical Representation

issue:

	dog	bite	man	is	a	pet	animal
(i) dog bite man	1	1	1	0	0	0	0
(ii) man bite dog	1	1	1	0	0	0	0
(iii) dog is a pet animal	1	0	0	1	1	1	1

context ସ୍ୱାତନ୍ତ୍ର୍ୟ ଓ sparse

↓ 0 वाक्य

Issue:

1. large + sparse matrices
2. completely ignores word order

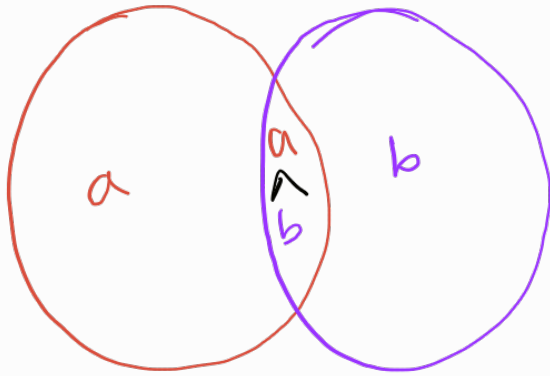
But often still useful

Classwork

[illegible]

Probability review

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$



prior

probability of a class before observing any data.

joint

both the data and the class occur together
multiple events happen at the same time.

$$P(A, B) \text{ or } P(A \wedge B)$$

conditional

one event given that another event is known.

Conditional probability,

$$P(A|B) = \frac{P(A \& B \text{ happens together})}{P(B \text{ happens})}$$
$$= \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A|B) P(B) = P(A \cap B)$$

$$\rightarrow P(A \cap B) = P(A|B) P(B) \rightarrow \text{Product Rule}$$

$$P(A \cap B | C) = P(A|C) \times P(B|C)$$

Conditional probability

→ A given B

$$P(A \cap B) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

Bayes Rule:

product rule:

$$P(A \cap B) = P(A|B) P(B) \quad \text{--- ①}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Swapping A and B,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(B|A) = \frac{P(A|B) P(B)}{P(A)}$$

 ← Bayes Rule

$$P(\text{spam} | \text{you won jackpot}) = \frac{P(\text{you won jackpot} | \text{spam}) P(\text{spam})}{P(\text{you won jackpot})}$$

spam file (ଅର୍ଥାତ୍ check କରନ୍ତୁ)
ଅର୍ଥାତ୍ you won jackpot ଅଟେ

$$h(o) = \operatorname{argmax}_{y \in Y} P(y|o)$$

↓
index return
ଅର୍ଥାତ୍

$Y \rightarrow \{+, -\}$

$$\begin{cases} \rightarrow P(+|o) \\ \rightarrow P(-|o) \end{cases}$$

→ ଅର୍ଥାତ୍ max
ଅଟେ (ଅର୍ଥାତ୍)

$$\begin{array}{l} \rightarrow \frac{P(o|+)P(+)}{\cancel{P(o)}} \\ \rightarrow \frac{P(o|-)P(-)}{\cancel{P(o)}} \\ \downarrow \\ \text{object} \end{array}$$

but what's the ans,
So, denominator $P(o)$
ଆମ (ମାମୁ) ଆମ calculation
ରୁ ହଟିଯାଏ, matter ନାହିଁ

using chain rule $\rightarrow$ computationally complex

naive bayes $\rightarrow$ every feature is independent to each other

$$P(\text{you, won, jackpot} | y^+) = P(\text{you} | y^+) P(\text{won} | y^+) P(\text{jackpot} | y^+)$$

$$P(\text{word} | \text{class}) = \frac{\text{count of 'word' in that class}}{\text{count of all word in that class}}$$

Date: 18.02.2026

$h(o) = \operatorname{argmax}_{y \in Y} P(y) \prod_{i=1}^m P(f_i(o)|y)$

Training data:

	f_1	f_2	f_3	f_{Sorry}	f_U	f_{call}	f_{can}	f_{have}	f_{later}	f_{me}	f_{now}	f_{won}
+ Sorry I'll call later	0	1	1	1	0	1	0	0	1	0	0	0
+ U can call me now	0	0	0	0	1	1	1	0	0	1	1	0
- U have won call now!!	2	0	0	0	1	1	0	1	0	0	1	1

Testing data:

→ Sorry! U can not unsubscribe 1 0 0 1 1 0 1 0 0 0 0 0 0

$P(+|o) \propto P(+)\prod P(f_i|+)\prod P(f_{\text{Sorry}}|+)\prod P(f_U|+)\prod P(f_{\text{can}}|+)$

$P(-|o) \propto P(-)\prod P(f_i|-)\prod P(f_{\text{Sorry}}|-)\prod P(f_U|-)\prod P(f_{\text{can}}|-)$

Naive Bayes use ~~for~~ prediction → Testing Data
Spam or not?

Ans:

$$P(\text{word}|\text{class}) = \frac{\text{count}(\text{word}) \text{ in that class} + 1}{\text{count of all word in that class} + |V|}$$

↓
vocabulary

$$h(o) = \operatorname{argmax}_{y \in Y} P(y) \prod P(f_i(o)|y)$$

$y \in \{+, -\}$

$$P(+|o) \propto P(+)\prod P(f_i|+)\prod P(f_{\text{Sorry}}|+)\prod P(f_U|+)\prod P(f_{\text{can}}|+)$$

$$= \frac{2}{3} \times \frac{0+1}{10+12} \times \frac{1+1}{10+12} \times \frac{1+1}{10+12} \times \frac{1+1}{10+12}$$

↓
positive class &
total word

↓
number of
unique words in the
training data

$$= 0.000023$$

$$p(-|o) \propto p(-) p(!|-) p(sorry|-) p(u|-) p(can|-)$$

$$= \frac{1}{3} \times \frac{2+1}{7+12} \times \frac{0+1}{7+12} \times \frac{1+1}{7+12} \times \frac{0+1}{7+12}$$

↓
vocabulary=12

$$= 0.000015$$

$$h(o) = \text{true} \quad (\text{since } 0.000023 > 0.000015)$$

lexicon → extra count of positive, negative emotions
pre annotated list

naïve bayes vs bag of words

Refining features:

① Binarization:

→ Instead of using frequency, we only focus
if the word is present or not

→ Equiv. of removing duplicate words for each doc.

② Lexicon features:

23.02.26

$$p(\text{word}|\text{class}) = \frac{\text{count of that word in that class} + 1}{\text{count of all word in that class} + |V|}$$

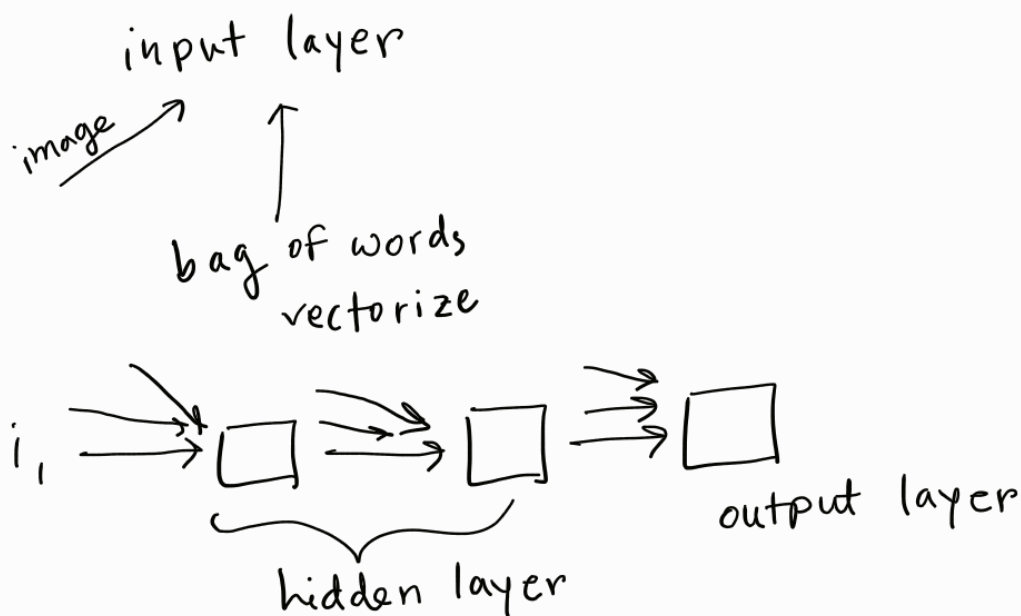
Concept of loss

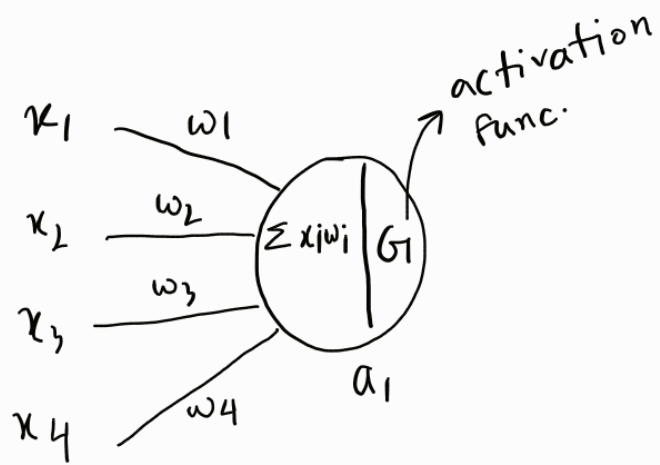
Binary Cross Entropy loss:

$$\text{Loss}_{\text{BCE}} = - [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

<u>Actual</u>	<u>Prediction</u>	$\text{Loss}_{\text{BCE}} = - [0 \cdot \log(0.1) + (1-0) \log(1-0.1)]$
0	0.1	$= -\log(0.9)$

Neural Network





$$a_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b$$

let, $w_0 = b$

$$x_0 = 1$$

then, $x_0 w_0 + x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4$

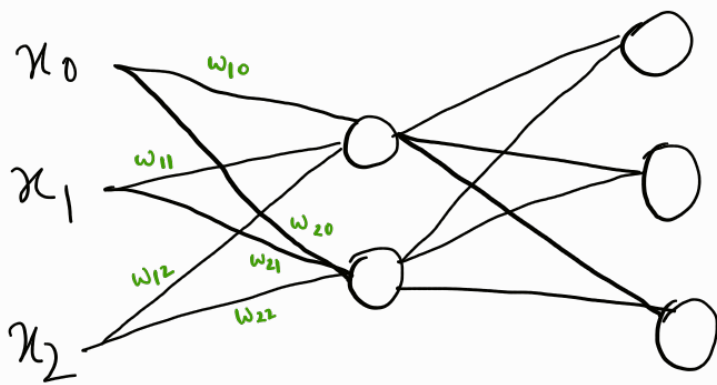
but why? → bcz $\sum$ matrix ৯ $\begin{bmatrix} x_0 & x_1 & x_2 & x_3 & x_4 \end{bmatrix}$ মানে

$$= \begin{bmatrix} x_0 & x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}$$

(1x5) (5x1)

$\times$ $\downarrow$

$$= G(X \cdot W)$$



$$\begin{matrix}
 \text{X} & W^{(1)} \\
 [x_0 \ x_1 \ x_2] & \begin{bmatrix} w_{10} & w_{20} \\ w_{11} & w_{21} \\ w_{12} & w_{22} \end{bmatrix} = [\sigma(a_1) \ \sigma(a_2)] \\
 [1 \times 3] & [3 \times 2]
 \end{matrix}$$

output of first hidden layer = $G(xw^{(1)})$

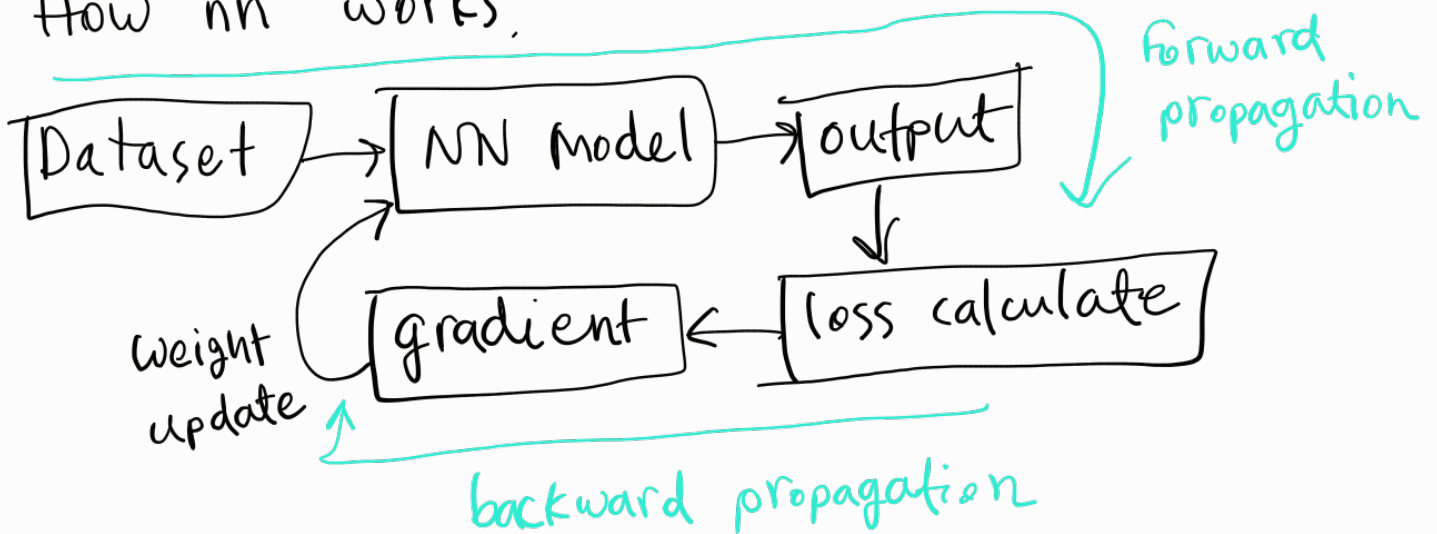
output of 2nd hidden layer = $G(G(xw^{(1)}) \cdot w^{(2)})$

Date: 25.02.26

Relu problem →

for nn a multiple activation function use
ଅନୁକ୍ରମେ ।

How nn works:



optimizer → weight optimize (gradient descent

stochastic gradient descent →
ଅବିଶିଷ୍ଟ data/dataset ଗୋଟିଏ ଗୋଟିଏ row
loss ଲାଗି
weight update ଲାଗି



mini-batch → କିଛି ଗୋଟିଏ ଗୋଟିଏ
(like 2^n ଲାଗି)

gradient descent → entire dataset
↓
learning rate fixed

Adaptive moment estimation $\rightarrow$ by default Adam
ଡାଲ୍‌ମାର Adam Adam

ADAM

underfitting $\rightarrow$ not complex enough

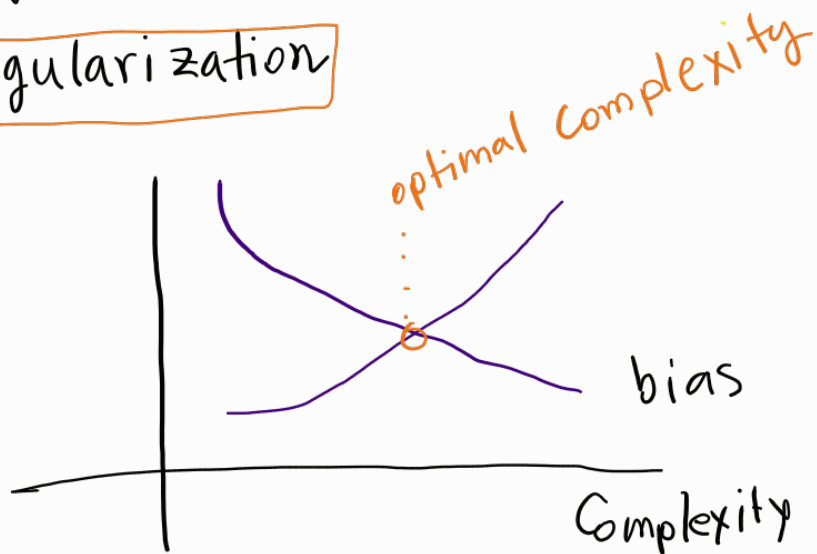
overfitting $\rightarrow$ over complex to learn
(noise fit fit fit fit fit)



Remove feature fit fit to make it simpler?



Regularization



Data splitting:

split 3 parts $\rightarrow$ Train
Validation
Test

Date: 02.03.26

Accuracy better matrix $\rightarrow$ imbalance issue

Confusion matrix

100 $\rightarrow$ 70 spam, 30 not spam

Model $\rightarrow$ 80 spam, 20 " "

20 not spam
Correct

	Predicted 0	Predicted 1
Actual 0	TN 20	FP 10
Actual 1	FN 0	TP 70

precision vs recall why:



False
positive

(email spam)



False negative

2 $\rightarrow$ combine $\rightarrow$ F1 score

100% accurate model means noise 3 (learn $\rightarrow$ $\rightarrow$)